Improved Knowledge Graph-Based Recommendation System

Enhancing Heterogeneous GNNs with Collaborative Signals on MovieLens 100K

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Motivation

- Traditional recommendation systems often suffer from **data sparsity** and **cold-start** problems.
- **Knowledge Graphs (KGs)** can integrate diverse types of side information (genres, user demographics) to improve representation learning.
- We propose an **Enhanced Heterogeneous Graph Neural Network** that not only leverages entity relations but also incorporates **collaborative filtering signals** (User-User, Movie-Movie similarities) directly into the graph structure.
- Our goal is to outperform standard baselines (Popularity, Content-Based) and provide **explainable** recommendations.

Methodology: Enhanced KG Model

Graph Construction:

- **Nodes:** User (943), Movie (1682), Genre (19), Occupation (21).
- **Edges:**
  - *Interaction:* User-Rates-Movie.
  - *Content:* Movie-Has-Genre.
  - *Profile:* User-Has-Occupation.
  - *Collaborative Signals:* User-Similar-To-User, Movie-Similar-To-Movie (using Cosine Similarity on interaction matrices).

Model Architecture:

- **Initialization:** Xavier Uniform for User, Movie, Genre, and Occupation embeddings (dim=128).
- **Convolution:** 3-layer Heterogeneous GNN.
  - Layer 1: **GraphSAGE** (Stability).
  - Layers 2-3: **GAT (Graph Attention Networks)** with 4 heads.
- **Regularization:** Layer Normalization, Dropout (0.2), Residual Connections.
- **Layer Combination:** Learned weighted summation of intermediate embeddings.
- **Link Predictor:** 3-layer MLP with Sigmoid activation.

Training Strategy:

- **Loss:** Combination of Bayesian Personalized Ranking (BPR) loss and InfoNCE contrastive loss.

$$\mathcal{L} = \mathcal{L}_{BPR} + 0.1 \cdot \mathcal{L}_{InfoNCE}$$

- **Negative Sampling:** Popularity-based sampling (power 0.75) with 4 negatives per positive.
- **Optimizer:** AdamW with ReduceLROnPlateau scheduler. Early stopping based on NDCG@10.

Experimental Setup

- **Dataset:** MovieLens 100K (90,570 train / 9,430 test ratings).
- **Baselines:**
  - *Popularity:* Recommend most frequently interacted items.
  - *Content-Based:* Cosine similarity on movie genres and user history.
- **Evaluation Metrics:** Recall@K, Precision@K, NDCG@K, Hit Rate@K (K=5, 10, 20).
- **Training:** 200 epochs, batch size 4096, learning rate 0.001.

Experimental Results

Table: Performance Comparison @10 and @20

Model	Recall	NDCG	Hit Rate	Recall@20
KG-GNN (Ours)	**0.1591**	**0.1647**	**0.7116**	0.2566
Popularity	0.1170	0.1213	0.6479	0.1847
ContentBased	0.0205	0.0202	0.1389	0.0356

The **KG-GNN** model significantly outperforms baselines, achieving a **+35.8%** NDCG improvement over the best baseline (Popularity).

Visual Analysis

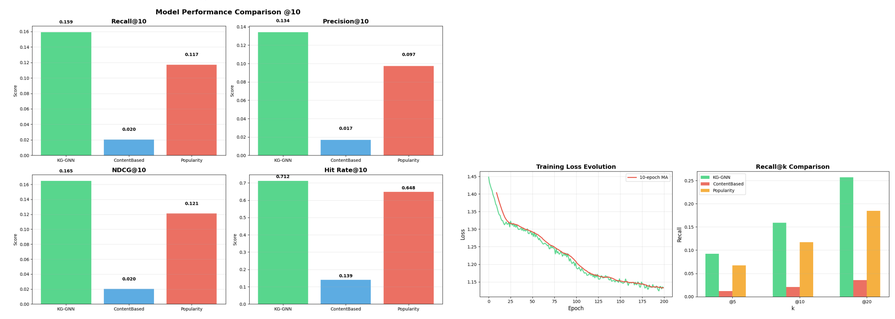

Figure: Left: Comprehensive comparison of metrics @10. Right: Recall@k across different models.

Figure: t-SNE visualization of Movie Embeddings colored by primary genre. The model successfully clusters semantically similar genres together.

Case Study: Explainable Recommendation

We analyzed the recommendation list for a sample user (ID 1, Occupation: technician) which included "The Hunt for Red October (1990)". The model leverages **Genre Affinities** (e.g., Action, Thriller) learned through the heterogeneous graph to bridge user preferences and movie content, even when explicit interaction history is sparse.

Embedding Quality Analysis

The KG structure effectively organizes the embedding space:

- **Within-Genre Similarity:** 0.4823
- **Between-Genre Similarity:** 0.3944
- **Separation Ratio:** 1.22x

Statistical t-test confirms a **highly significant** difference between intra- and inter-genre similarities ($p \approx 0$), validating the model's ability to learn meaningful content representations.

User Embedding Clustering

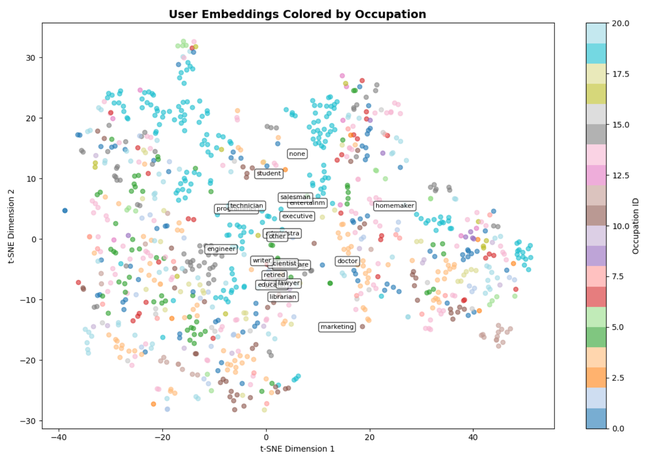

Figure: User Embeddings colored by Occupation. Distinct occupational groups form identifiable clusters (e.g., 'educator', 'programmer'), allowing the model to transfer preferences between similar users.

Conclusions & Future Work

Key Insights:

1. **Side Information Matters:** Integrating Genres and Occupations via a KG significantly enriches representations.
2. **Collaborative Signals are Crucial:** Adding User-User and Movie-Movie similarity edges boosts performance beyond content filtering.
3. **Explainability:** The graph structure allows tracing recommendation logic through genre affinities and user clusters.

Future Work:

- Incorporate **temporal dynamics** (timestamps) into the graph.
- Explore **multi-task learning** (simultaneous rating prediction and link prediction).
- Implement **attention-based explanation** mechanisms to highlight key nodes and edges.